UNIVERSITY OF SCIENCE & TECHNOLOGY OF HANOI
TRƯỜNG ĐẠI HỌC KHOA HỌC & CÔNG NGHỆ HÀ NỘI

GROUP PROJECT TOPIC PROPOSAL

Specialty: - Information and Communication Technology
- Data Science
Academic year: 2026 - 2027
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Topic: Few-shot Learning in Blood cancer cell Detection

Description:
- To detect blood cancer, doctors must navigate a complex diagnostic and testing process that demands extensive training and years of clinical experience. Even so, diagnostic discrepancies can still arise depending on an individual practitioner's skill and expertise. In resource-limited medical centers where specialized staff is scarce, these challenges can directly impact patient outcomes and even cost lives.
- This project trains a computer system to assist in this screening stage by localizing and flagging suspicious blood cells for clinician review. Instead of relying on tens of thousands of expert-labeled images which are prohibitively expensive and difficult to obtain in the medical domain that the system learns from only a handful of examples per cell type, much like a medical student learning to recognize a pathology after seeing just a few reference cases. Furthermore, we evaluate the system's robustness across images from different sources.

Expected outcomes:
- A working prototype that classifies blood cells as normal or abnormal with a confidence score, along with heatmaps showing which part of the cell the model relied on. We will also release open-source code and a standard evaluation protocol, plus a technical report showing how many examples the system needs and how well it holds up on data from other sources. If time allows, a demo will scan a full blood slide and mark suspicious cells. This is a research prototype that supports screening, not a diagnostic tool.

Used Methods and Techniques:
- We train a computer to find and recognize blood cancer cells from only a few labeled examples, using vision-only models. A YOLO detector first learns to locate white blood cells in microscope images, and a self-supervised vision foundation model, DINOv2, learns general cell features from unlabeled images without any text. We then compare three ways to adapt it to leukemia: fine-tuning only a small part of the network with LoRA, Prototypical Networks that classify cells by their similarity to a few labeled examples, and an example-based foundation detector (DE-ViT) that finds new cell types from just a handful of reference images. Stain normalization and color augmentation help it cope with images from different microscopes and datasets, which we test by training on one dataset and evaluating on another. Results are averaged over hundreds of random trials and measured by mAP, sensitivity, specificity, and F1, with heatmaps showing what the model focused on.

Leader's signature **Supervisor's signature**

Department confirmation **Director of ICT department**

- Approved
- Need to consider
- Other comments